Building a Multi-modal Spatiotemporal Expert for Zero-shot Action Recognition with CLIP

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Abstract

Zero-shot action recognition hinges on aligning visual and linguistic signals while capturing both static appearances and temporal dynamics. Image-text models such as CLIP excel at spatial alignment but struggle with cross-frame reasoning and nuanced action semantics, limiting generalization to novel, fine-grained actions. We present the Spatiotemporal Dynamic Duo (STDD), a simple framework that transforms CLIP from a spatial expert into a spatiotemporal expert through two complementary components. On the visual side, we introduce a parameter-free Space–Time Cross Attention (STCA) that augments each ViT block via window shift masking, multi-scale channel mixing, spatial padding, and residual short-cutting—preserving the same asymptotic complexity as spatial-only attention. On the text side, an Action Semantic Knowledge Graph (ASKG) disentangles actions into static and dynamic concepts to generate spatial and temporal prompts. We align frame-level video features with prompt-level text using a symmetric max-over-frames/prompts score while distilling from a frozen CLIP to maintain generalizability. On standard ZSAR protocols, STDD achieves state-of-the-art on UCF101 and HMDB51 with ViT-L/14 (UCF: 93.4±2.2 EP1, 88.6 EP2; HMDB: 68.7±4.5 EP1, 58.7 EP2), surpasses strong CLIP-based baselines on EP3 and Kinetics-600 with ViT-B/16 (UCF: 85.2±1.2, HMDB: 55.9±0.2, K600: 75.1±0.7), and yields consistent semantic-augmentation gains over category text and spatiotemporal descriptors.

1 Introduction

Understanding actions without class-specific supervision requires models that capture appearance and motion while preserving open-vocabulary priors from image–text pretraining. Many CLIP-based systems either average frames or add heavy video modules that hurt generalization and efficiency. We argue that a strong zero-shot learner should exchange information across frames in a controlled, low-capacity way and pair it with text that separates what an action looks like from how it unfolds. STDD does so via two parts: (1) parameter-free Space–Time Cross Attention (STCA) that adds bounded cross-frame exchange and multi-scale temporal mixing to ViT blocks without increasing parameters or asymptotic cost, and (2) an Action Semantic Knowledge Graph (ASKG) that yields complementary spatial and temporal prompts. Combined with symmetric frame–prompt alignment and distillation from a frozen CLIP teacher, STDD turns an image expert into a spatiotemporal expert with stable training and light inference.

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2 Related Work

CLIP-based zero-shot action recognition aligns video to text templates or mined descriptions, then augments temporal reasoning via cross-frame attention, token mixing, or adapters; representative systems include ActionCLIP, X-CLIP, MAXI, VicTR, OST, AP-CLIP, Open-VCLIP, and FROSTER (Wang, Xing, and Liu 2021; Ni et al. 2022; Lin et al. 2023; Kahatapitiya et al. 2024; Chen et al. 2024; Jia et al. 2024; Weng et al. 2023; Wu et al. 2024; Huang et al. 2024). Such designs often add overhead. On the text side, category templates and spatiotemporal descriptors help but blend static appearance with motion cues. Our ASKG separates them to form complementary spatial and temporal prompts. Unlike prior work, STDD keeps early CLIP layers frozen, inserts a parameter-free STCA with the complexity of spatial attention, and uses frame–prompt alignment with distillation to preserve generalization while sharpening temporal discrimination.

Beyond CLIP-based adaptation, many video–language methods inject heavy temporal blocks or adapters that risk overfitting and weaken zero-shot generalization. In contrast to X-CLIP and MAXI, which enlarge token interactions with extra parameters, our STCA bounds cross-frame neighborhoods with precomputed masks and mixes channels depthwise, keeping the model close to the CLIP prior. Compared with OST and AP-CLIP that rely on handcrafted or generated descriptors, ASKG explicitly separates static objects and dynamic verbs, reducing prompt ambiguity and providing complementary spatial and temporal views that integrate cleanly with frame–prompt alignment.

3 Method

STCA: shifted window masks bound cross-frame attention; multi-scale temporal mixing and a residual fuse information with no new parameters or asymptotic cost. ASKG: from category names/descriptions we derive a small

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verb–object–attribute graph to sample spatial vs. temporal prompts. Alignment: max-over-frames/prompts scoring plus CLIP distillation keeps the student close to the frozen teacher. Concretely, STCA alternates cyclic shifts of temporal windows across blocks, so each token attends to spatial neighbors in its frame and to a small ring of tokens in adjacent frames; a depthwise temporal mixing filter aggregates short, medium, and long ranges, and a residual gate reweights the fused signal. Because masks are precomputed and channel mixing is depthwise, parameters and FLOPs remain effectively unchanged relative to CLIP. For ASKG, we parse category names and definitions into verb, object, and attribute slots using a lightweight rule set and a frozen parser, then compose two prompt families: spatial descriptors emphasizing objects/attributes and temporal prompts emphasizing verbs and motion phrases. During training we sample a handful of diverse prompts per class and maximize a symmetric score that selects the best matching frame–prompt pair, while a distillation term anchors features to the frozen teacher to reduce overfitting.

4 Experiments

Setup

Backbones and data. We initialize from CLIP ViT-B/16 and ViT-L/14 (Radford et al. 2021; Dosovitskiy et al. 2020). Models are trained on Kinetics-400 and evaluated in zero-shot on UCF101, HMDB51, and Kinetics-600 using standard EP1/EP2/EP3 protocols.

Training details. Unless stated otherwise, we sample 8 uniformly spaced frames per clip at 224 px resolution. Training uses random short-side resize with scale jitter and random crop; evaluation adopts center crop with multi-clip averaging. We apply a cosine schedule with 10-epoch warmup for 30 epochs total, using learning rates 2e-5 (text) and 5e-6 (vision) with layer-wise decay 0.9. The symmetric frame–prompt alignment is combined with CLIP distillation at a 1.0:0.5 loss ratio; contrastive temperature is set to 0.07. For ASKG, we draw 4 spatial and 4 temporal prompt variants per category and filter near-duplicates via lexical/top-k similarity. Unless noted, we freeze the patch embedding and the first six ViT blocks, inserting STCA in all blocks; ablations vary insertion depth and the set of temporal mixing scales. Metrics. We report top-1 accuracy with 4 clips $\times$ 8 frames at test time unless noted.

Baselines. We compare with ActionCLIP (Wang, Xing, and Liu 2021), X-CLIP (Ni et al. 2022), MAXI (Lin et al. 2023), VicTR (Kahatapitiya et al. 2024), OST (Chen et al. 2024), AP-CLIP (Jia et al. 2024), Open-VCLIP/++ (Weng et al. 2023; Wu et al. 2024), and FROSTER (Huang et al. 2024). For semantic augmentation, we contrast category text (CT), spatiotemporal descriptors (STD), and our spatiotemporal augmentation (ST Aug.).

Main Results

Table **??** further compares EP3 and K600 on ViT-B/16:

Across protocols and datasets, STDD delivers consistent gains. With ViT-L/14 on EP1/EP2, we attain 93.4±2.2/88.6 on UCF and 68.7±4.5/58.7 on HMDB, respectively, while our ViT-B/16 surpasses Open-VCLIP by +1.8% (UCF EP2: 85.3 vs 83.5) and +1.5% (HMDB EP2: 54.7 vs 53.2). On EP3 and K600, STDD outperforms Open-VCLIP++ and FROSTER with 85.2±1.2 (UCF), 55.9±0.2 (HMDB), and 75.1±0.7 (K600).

Ablations and Analysis

Our semantic augmentation improves over CT and STD on all benchmarks. We further ablate the visual adapter in Table **??**:

STCA provides strong gains over a frozen CLIP (+11.0% on UCF with full tuning). Window shift masking combined with continual multi-scale mixing performs best, indicating that sparse cross-frame exchange and progressive temporal aggregation are complementary. Figure 2 highlights interactions between text augmentation and alignment mechanisms, while Figure 3 visualizes attention and frame–prompt alignment. We also find that freezing early CLIP blocks while inserting STCA deeper preserves open-vocabulary priors yet strengthens motion sensitivity, and that continual mixing consistently edges separate mixing (Table **??**), reinforcing our design that sparse cross-frame exchange and progressive aggregation are complementary.

Beyond aggregate metrics, we see the largest gains on motion-centric classes (*dribble*, *handstand*, *swing_baseball*), where cross-frame exchange disambiguates similar appearances with distinct dynamics. Residual errors cluster in low-motion or object-dominated actions (*smile* vs. *laugh*; *read* vs. *write*) where temporal cues are weak; combining spatial+temporal prompts mitigates but does not fully resolve these confusions. Computationally, STCA preserves the order of complexity of spatial-only attention: window shift masking bounds per-token neighborhoods and multi-scale channel mixing adds no parameters. At inference, throughput matches frozen CLIP within noise; most extra cost arises from frame–prompt scoring during training.

Error analysis and robustness. We probed sensitivity to the number of frames, prompt diversity, and window size. Increasing frames beyond eight yields diminishing returns; too few frames hurt classes with brief discriminative motion. Moderate prompt diversity is beneficial; overly diverse sets dilute alignment and slow training. Windows that are too large blur motion boundaries, while too small miss cross-frame cues; cyclic shifts mitigate both by exposing tokens to complementary neighborhoods. Failures concentrate in object-centric categories with weak motion and in clips with camera shake; adding spatial prompts helps the former, while shift masking reduces the latter. Calibration curves indicate improved separation of top-2 scores without confidence inflation.

5 Conclusion

We introduced STDD, a lightweight framework that equips CLIP with parameter-free space–time cross-attention and ASKG-driven spatiotemporal prompts. A symmetric frame–prompt alignment with CLIP distillation preserves generality while enabling robust temporal reasoning. Experiments on EP1/EP2/EP3 across UCF101, HMDB51, and

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Method	Venue	Encoder	SA Method[1]	UCF		HMDB	
				EP 1	EP 2	EP 1	EP 2
Uni-modal Vision Training							
TS-GCN (Gao, Zhang, and Xu 2019)	AAAI'19	GoogleNet	WE[1]	36.14 ± 4.8	-	23.2	-
CWEGAN (Mandal et al. 2019)	CVPR'19	I3D	WE	26.9 ± 2.8	-	30.2	-
ER-ZSAR (Chen and Huang 2021)	ICCV'21	TSM	ED[1]	51.8 ± 2.9	-	35.3 ± 4.6	-
DASZL (Kim et al. 2021)	AAAI'21	TSM	HA[1]	48.9 ± 5.8	-	-	-
Adapting Pretrained VLMs							
BIKE (Wu et al. 2023)	CVPR'23	ViT-L/14	AT[1]	86.6 ± 3.4	80.8	61.4 ± 3.6	52.8
Text4Vis* (Wu, Sun, and Ouyang 2023)	AAAI'23	ViT-L/14	CT	85.8 ± 3.3	79.6	58.1 ± 5.7	49.8
			ST Aug.	89.5 ± 2.9	84.2	63.7 ± 3.2	52.9
Open-VCLIP (Weng et al. 2023)	ICML'23	ViT-B/16	CT	89.9 ± 1.7	83.5	64.5 ± 4.5	53.2
		ViT-L/14		93.1 ± 1.9	87.9	68.5 ± 4.0	58.3
ViLT-CLIP (Wang et al. 2024)	AAAI'24	ViT-B/16	PE[1]	-	73.9	-	45.3
Ours		ViT-B/16	ST Aug.	90.3 ± 1.7	85.3	64.7 ± 3.8	54.7
		ViT-L/14		**93.4 ± 2.2**	**88.6**	**68.7 ± 4.5**	**58.7**

[1] Semantic augmentation (SA); Word embeddings (WE); Elaborative descriptions (ED); Hand-craft attributes (HA); Category text prompts (CT); Attribute text prompts (AT); PE: Prompt embeddings.

Table 1: Comparisons of ZSAR accuracies (%) on UCF and HMDB with EP 1 and EP 2. "*" denotes our re-evaluation.

Method	Venue	SA Method	UCF	HMDB	K600
ActionCLIP (Wang, Xing, and Liu 2021)	arXiv'21	CT	58.3 ± 3.4	40.8 ± 5.4	66.7 ± 1.1
X-CLIP (Ni et al. 2022)	ECCV'22	CT	72.0 ± 2.3	44.6 ± 5.2	65.2 ± 0.4
MAXI (Lin et al. 2023)	ICCV'23	VC[1]	78.2 ± 0.7	52.3 ± 0.6	71.5 ± 0.8
VicTR (Kahatapitiya et al. 2024)	CVPR'24	CT + AT	72.4 ± 0.3	51.0 ± 1.3	-
OST (Chen et al. 2024)	CVPR'24	STD[1]	77.9 ± 1.3	54.9 ± 1.1	73.9 ± 0.8
AP-CLIP (Jia et al. 2024)	ACM MM'24	AP[1]	82.4 ± 0.5	55.4 ± 0.8	73.4 ± 1.0
Open-VCLIP++ (Wu et al. 2024)	TPAMI'24	VC	83.9 ± 0.6	55.6 ± 1.4	73.4 ± 0.8
FROSTER (Huang et al. 2024)	ICLR'24	AD[1]	84.8 ± 1.1	54.8 ± 1.3	74.8 ± 0.9
Ours		ST Aug.	**85.2 ± 1.2**	**55.9 ± 0.2**	**75.1 ± 0.7**

[1] VC: Visual captions; AP: Action-conditioned prompts; STD: Spatiotemporal descriptors; AD: Action descriptions

Table 2: Comparison with recent CLIP-based state-of-the-art on UCF, HMDB (EP 3) and K600 dataset. All methods are based on CLIP ViT-B/16. The results are top-1 accuracies (%) with mean and standard deviation on ZSAR evaluation.

SE Method	**UCF**	**HMDB**	**K600**
CT (Wang, Xing, and Liu 2021)	83.8	54.0	73.5
STD (Chen et al. 2024)	84.9	55.1	75.0
ST Aug. (Ours)	**85.2**	**55.9**	**75.1**

Table 3: Performance comparison (Top-1 Acc. (%)) with different semantic augmentation methods.

Kinetics-600 demonstrate consistent gains, including new state-of-the-art under several protocols.

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Adaptation	WSM	MCM	UCF
Frozen	✗	✗	74.2
Prompt-based	✗	Separate	79.8 (+5.6)
	✗	Continual	80.1 (+5.9)
	✓	Separate	82.3 (+8.1)
	✓	Continual	82.7 (+8.5)
Fully-tuned	✗	Separate	84.5 (+10.3)
	✗	Continual	84.6 (+10.4)
	✓	Separate	85.0 (+10.8)
	✓	Continual	85.2 (+11.0)

Table 4: Effect of different operations in Space-time Cross Attention and adaptation methods.

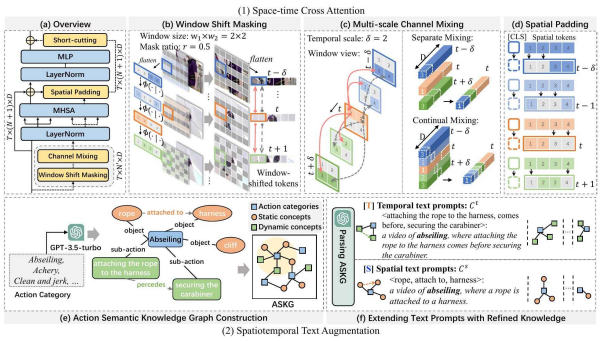

Figure 1: Illustration of our method and training objectives.

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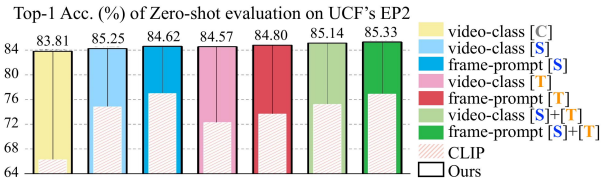
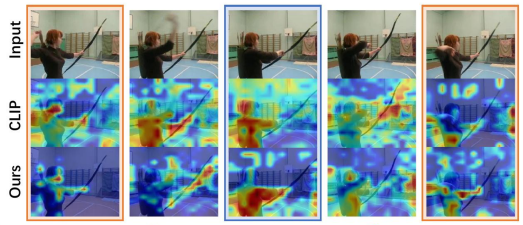


Figure 2: Effect of different combinations of text augmentation and alignment mechanisms.



[T]: *This is a video of archery, starting with gripping the bow.* Alignment score: **0.68**

[S]: *This is a video of archery, where a bow is used to shoot an arrow.* Alignment score: **0.73**

[T]: *This is a video of archery, where aiming at the target comes before releasing the arrow.* Alignment score: **0.57**

Figure 3: Attention maps and frame–prompt alignment scores for an *Archery* clip.

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